

Final Report

Tethersonde: Meteorological Observation Device

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Executive Summary

This report outlines the final findings of our senior design project to enhance the tethersonde, a vital meteorological observation device utilized by the Environmental Fluid Dynamics Lab (EFD) at the University of Utah. The primary objective was to augment the existing tethersonde design with real-time data transmission, wind speed, direction measurement, and a battery life indicator while retaining the original sensor array.

To achieve these goals, our approach involved integrating a thermal anemometer for air speed measurement, an IMU (Inertial Measurement Unit) for azimuthal direction, a LoRaWAN-capable microcontroller for communication and computing, and a large RGB LED for status indication. Adherence to EFD's specifications, including wind speed measurement within the range of 0.5-10 m/s with a resolution of 0.1 m/s, azimuthal measurement within 5°, operating height of 1000 meters, the minimum sampling rate of 1Hz, weight under 400 grams, and operational duration of 2+hours. These specifications were ensured throughout the project.

Key components such as the Modern Device Rev.P wind sensor, BNO085 9-DoF IMU, and the Arduino MKRWAN 1310 microcontroller were utilized to meet these specifications. A comprehensive analysis encompassing power spectrum analysis, circuitry verification, and performance evaluation under transient conditions, mainly focusing on the capabilities of the Modern Device Rev.P wind sensor, was conducted.

In conclusion, our enhancements to the tethersonde have successfully met EFD's specifications, providing improved functionality for atmospheric boundary layer analysis while maintaining the integrity of the original sensor array.

Design Metrics

Table 1: Design requirements for a singular sonde

Design Requirements	Expectations	Reality
Weight of Sonde	<400 grams	315 grams
Wind Speed Accuracy	± 0.1 m/s	± 0.2 m/s
Azimuthal Measurement Accuracy	$<5^\circ$	0.433°
Communication Protocol Range	$\cong 1000$ m	1000 m
Operational Duration	2+Hours	4 Hours
Sampling Rate	1Hz	5Hz

Design Concepts

Tethersondes are invaluable tools for analyzing the boundary layer of the Earth's atmosphere. By measuring temperature, humidity, pressure, and velocity at various altitudes, tethersondes provide crucial data on the vertical structure of the boundary layer, helping researchers understand atmospheric turbulence, heat transfer, and flow patterns. Tethersondes are deployed on moored balloons instead of disposable high-altitude balloons, creating a fixed testing environment for measurements and extended-duration tests at fixed altitudes.



Figure 1: The Vaisala tethersonde was previously used for boundary layer analysis, but its equipment has become outdated and lacks customizability. The EFD lab desired a tethersonde they could produce themselves that would perform at or above the level of Vaisala. This company created a winch and balloon while boasting a maximum of six sondes that could be used at a time. It utilized a cup and vane-style anemometer, which made the device heavy and lacked turbulence resolution.

The EFD lab uses tethersondes for a boundary layer analysis, whether they are looking at flows in the desert or an urban environment. There are few commercial tethersondes in production, and those that exist are outdated and require proprietary equipment that isn't compatible with most modern computers (Fig.1). The EFD lab began to develop its tethersonde, which could be produced using campus resources and easily replaceable sensors. Their design included a Wi-Fi transceiver-capable microcontroller, a barometer, an SD card breakout, thermometers, and a hygrometer. The prototype was developed with a rushed timeframe for its production, which didn't allow for much experimentation to solve the problem of gathering airspeed and azimuthal direction. The sensors included onboard the sonde, the SHT47, BMP390, and E-type thermocouple were well-documented and capable of resolution and acceptable error within the scope of the tethersonde application. The EFD lab desired a solution to the airspeed and direction sensors, battery life indication, and remote system control from a base station.

The EFD lab's specifications are to measure wind speed in the 0.5-10 m/s range with a resolution of 0.1 m/s, azimuthal measurement within 5° accuracy, an operating height of 1000m, and a minimum sampling rate of 1Hz. The sensors and circuit additions to complete these specifications are the Modern device Rev. P wind sensor, BNO085 9-Dof IMU, and the Arduino

MKRWAN 1310 LoRa WAN capable microcontroller. TPS61023 5V and TPS61040 12V boosters were added to achieve the necessary power for the additional components. The Rev. P is a lightweight constant temperature thermal anemometer that works on a hot wire principle (EQ.1). These devices can be very accurate at low air speeds, their weight and size are unmatched by any anemometer, and they have an analog output for variable voltage which can be calibrated to airspeed. The IMU allows proper heading and attitude measurements to determine the azimuthal angle while stable depending on the AHRS (attitude heading reference system) filtering method. The original microcontroller used by the EFD lab was sufficient to run all the circuitry; however, the Arduino selected was of similar size, LoRa WAN capable, and its analog pins were capable of 12-bit resolution.

Final Design Description

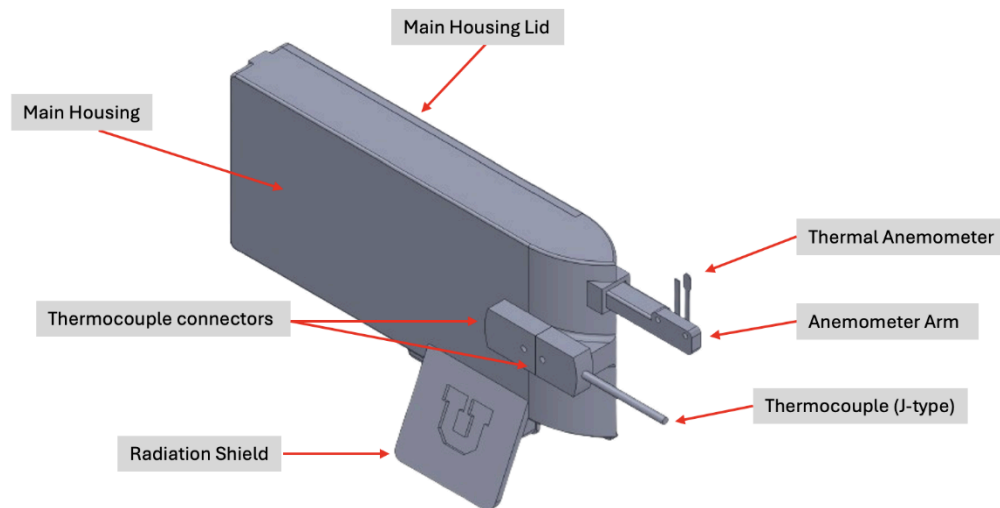


Figure 2: Tethersonde outer body assembly. The anemometer arm is on the bow, while the lid is on the port side of the frame. The thermocouple sits on the starboard side of the device.

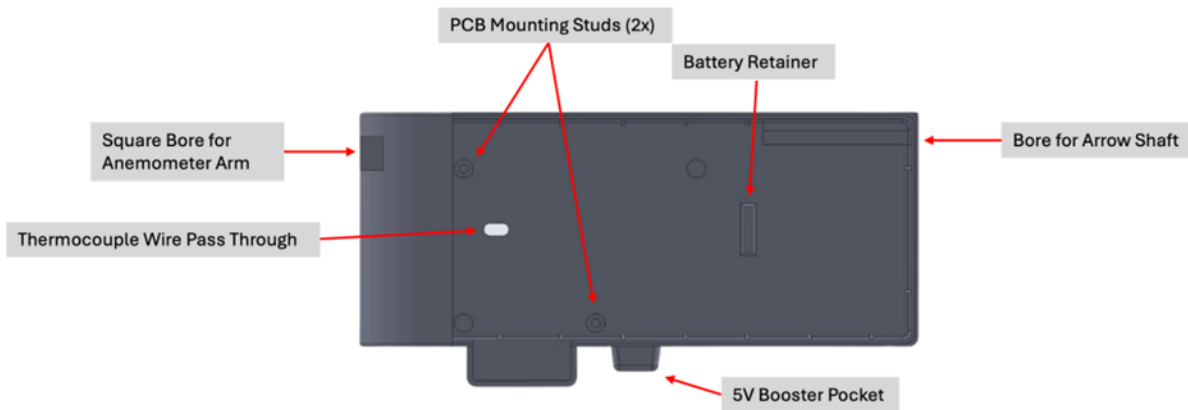


Figure 3: View of the port side of the tethersonde without the lid. Inside the PCB mountain points, wire pass-throughs, anemometer arm bore, and arrow shaft bore are displayed.

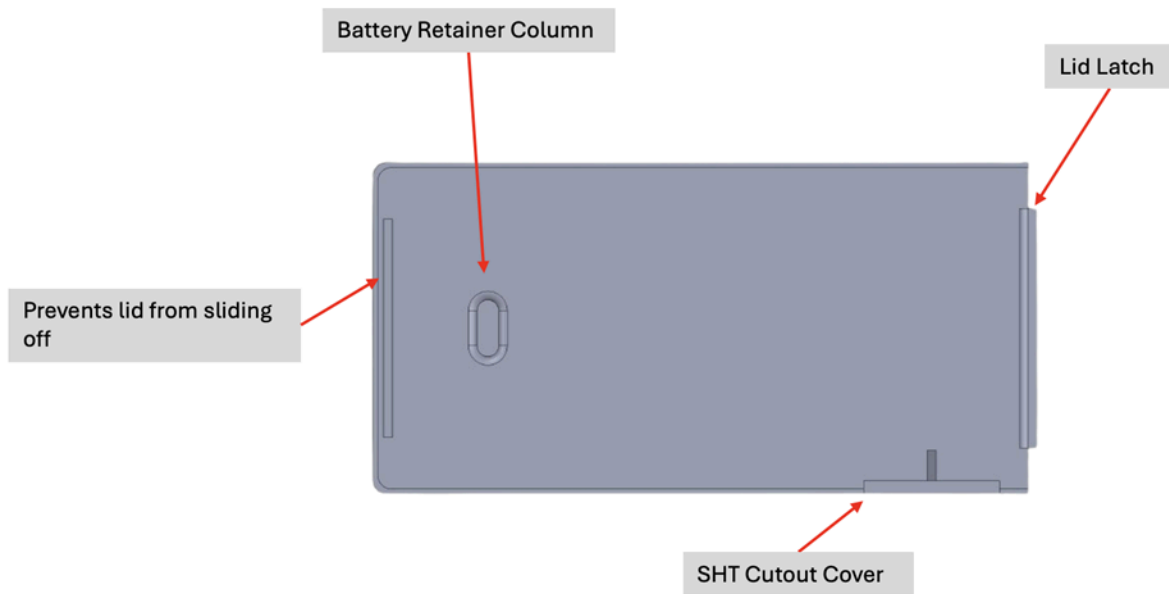


Figure 4: The internal side of the lid. This image shows the attachment points and the battery retention column, which keeps everything within the sonde restrained from movement.

Table 2: Cost of parts per sonde. Each part in the spreadsheet is onboarded to the sonde, including the manufacturing costs such as 3D printing and outsourcing the PCB production to OshPark.

Part/ Part#	Description	Unit Cost
Arduino MKR WAN 1310	Microcontroller	\$45.60
BMP390	Barometer	\$10.95
MAX31856	Thermocouple Amplifier	\$17.50
SHT45	Temperature/hydrogrometer	\$12.50
MicroSD	Data logger	\$7.50
BN0085	IMU	\$24.95
Wind Sensor Rev P	Thermal anemometer	\$39.95
Li-Po- 3.7v 2500mAh	Rechargeable Battery	\$12.50

Table 3: The total budget for the project. Funding included FATIMA project funding allocated to us by the EFD lab and the budget allotted by the Senior design course. The lab will utilize the leftover budget to either produce more sondes or adjust the current models with the recommendations stipulated in this document.

Part Identifier	Description	Vendor	Quantity	Unit Price	Total Price	Received
0.1	Arduino MKR WAN 1310	Arduino	7	\$45.60	\$319.20	7
0.2	BMP390	Adafruit	6	\$10.95	\$65.70	3
0.3	MAX31856	Adafruit	6	\$17.50	\$105.00	3
0.4	SHT45	Adafruit	6	\$12.50	\$75.00	3
0.5	MicroSD	Adafruit	6	\$7.50	\$45.00	3
0.6	BN0085	Adafruit	6	\$24.95	\$149.70	3
0.7	Wind Sensor Rev P	Modern Device	8	\$39.95	\$319.60	4
0.8	Li-Po- 3.7v 2500mAh	Adafruit	6	\$12.50	\$75.00	3
0.9	QT to QT cable (50mm)	Adafruit	15	\$0.95	\$14.25	15
1	QT to QT cable (100mm)	Adafruit	0	\$0.95	\$0.00	NA
1.1	Bread board		1	\$0 (borrowed)	\$0.00	NA
1.2	Jumper Wire Kit		1	\$0 (borrowed)	\$0.00	NA
1.3	UCF Antenna	Arduino	4	\$5.60	\$22.40	3
1.4	RGB light	Adafruit	3	\$2.00	\$6.00	NA
1.5	TPS61023 5v Booster	Adafruit	11	\$3.95	\$43.45	3
1.6	TPS61040 12v Booster	Adafruit	8	\$2.95	\$23.60	3
1.7	JST-PH 2-Pin Connector	Adafruit	13	\$0.75	\$9.75	3
1.8	Custom PCB	Osh Park	6	\$17.00	\$102.00	3
1.9	JST conn header smd 5pos 1mm	Digikey	18	\$0.72	\$12.96	6
2	Lora Gateway Kit	Digikey	1	\$328.00	\$328.00	1
2.1	QT-QT Stemma extension	Digikey	8	\$1.73	\$13.84	3
2.2	Lora Antenna		5	\$2.95	\$14.75	5
2.3	Battery Extension	Adafruit	5	\$1.95	\$9.75	5
2.4	Headers/Fasteners/Resistors		1	\$2.00	\$2.00	1
2.5	RTC	Adafruit	6	\$13.50	\$81.00	NR
				Total:	\$1,837.95	
				Budget:	\$2,500	
				Remaining	\$662.05	

Engineering Analysis

When selecting the additional sensors to meet the specifications given by the EFD lab, the priority was to find small and easily sourced sensors. The EFD lab's prototype sensors were all sourced from Adafruit, which inclined the senior design team to look at other sensors in their suite. Adafruit provided good documentation for all the sensors, which led the team to discover an Inertial Measurement Unit (IMU) that could be used to determine the azimuth component of the project.

With the senior design team's initial experience with microcontrollers, an Arduino microcontroller, the MKRWAN1310, was selected. This Arduino was explicitly chosen for its size, features, 12-bit analog pins, and LORAWAN. This microcontroller provided the necessary features to meet the user specifications.

For the wind sensor component of the Tethersonde, the senior design team explored two ideas: a thermal anemometer and a pitot tube. The initial analysis of the pitot tube determined

that the pressure transducer's size required to meet the specifications would be far too large and heavy. The thermal anemometer, RevP. It had very little documentation, but a research paper from RMIT University in Melbourne, Australia, provided insights into the performance of the RevP and how the senior design team could improve upon it. The RevP needed to align with King's law, a heat transfer relationship where the electrical power is equal to the heat removed from the exposed heating element due to convection. The convective heat transfer coefficient contains a velocity component that relates the sensor's voltage output to the surrounding air's velocity (EQ1).

Once the suite of additional sensors was selected, the senior design team needed to verify that each sensor could meet the specifications. The IMU performance must be verified by testing it against magnetic north to prove it could be used like a compass. The LoRa chip on the microcontroller needed verification on the range as none of the team or advisors had previously worked with the LoRa communication protocol. The RevP required the most verification as there was very little documentation from the company, and only one published research paper about its older model existed.

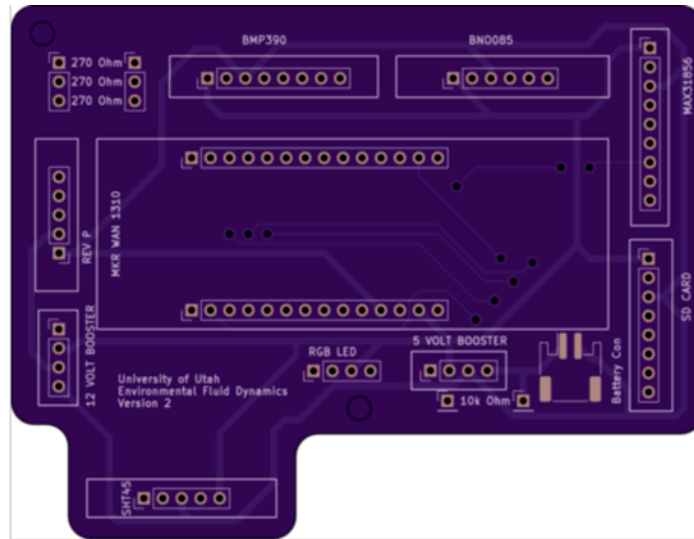


Figure 5: This PCB was designed for the tetheredsonde. It connected all the necessary sensors and voltage boosters to the microcontroller, and it was intended not only for prototyping and verification but also for the final design.

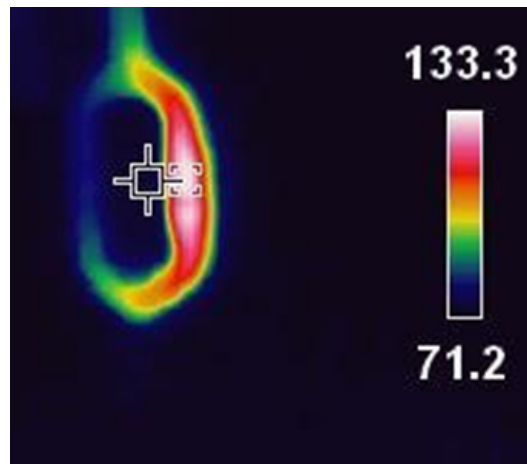


Figure 6: This is a thermal image of the RevP being powered with 12V. It clearly shows the heat output across the RevP's thermistor, which serves as this thermal anemometer's "Hot-wire" component. Thermal images were taken to verify that the same heat output was achieved from multiple power sources, such as the USB power from the microcontroller or the Lipo battery powering the entire PCB.

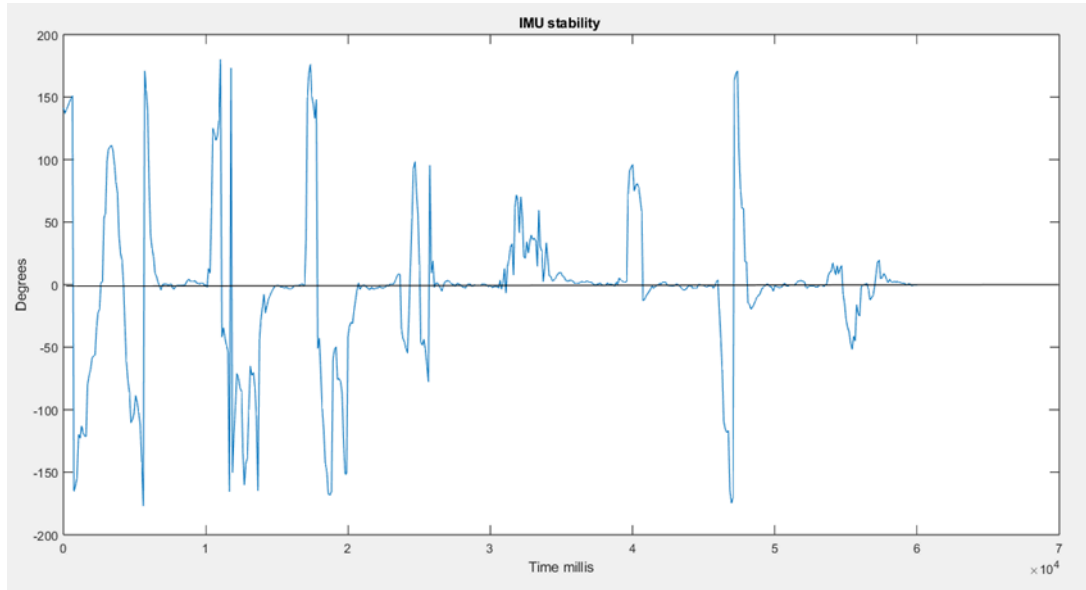


Figure 7: Imu stability on its magnetic north bearing was a significant part of the proof of concept. If the IMU couldn't reliably detect magnetic north, then the IMU wouldn't have been able to be used on board the sonde. This plot shows how agitations applied to the IMU did not affect its ability to orient itself to magnetic north, the 0-degree line for this figure. More tests were required to determine the IMU's accuracy and response time, but this initial test proved that the Imu could accurately detect azimuthal components such as North, South, East, and West.

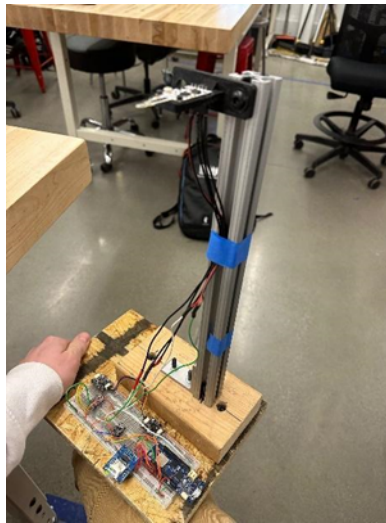


Figure 8: Testing frame for RevP. Initially, the plan was to calibrate the RevP at high windspeeds, 1 – 10 m/s, in the wind tunnel. This calibration would serve as a starting point for calibrating low-speed values, 0 - 1m/s, which would be performed outside against the sonic anemometer. Depicted here is the RevP on a 3D printed mount, which was elevated by an aluminum T-slot part with the accompanying microcontroller that would log the voltage output from the RevP. The RevP was tested against a pitot static tube built into the wind tunnel for steady-state values.

Table 4: Steady-state testing values gathered from the wind tunnel tests. Using the modified Bernoulli's equation (EQ2), airspeed values can be collected from the pressure transducer values. The steady-state test generated a 2-minute average voltage value at every airspeed measured in the wind tunnel.

Test Number	Pressure Transducer Delta (mmHg)	Airspeed from Pitot tube (m/s)	Average Voltage output from Rev.P
10	0.415	10.69	0.8506
9	0.353	9.91	0.8317
8	0.271	8.77	0.8058
7	0.184	7.37	0.7708
6	0.146	6.67	0.7504
5	0.078	5.17	0.7028
4	0.045	4.27	0.6695
3	0.019	3.38	0.6254
2	-0.002	2.44	0.5707
1	-0.016	1.53	0.4949

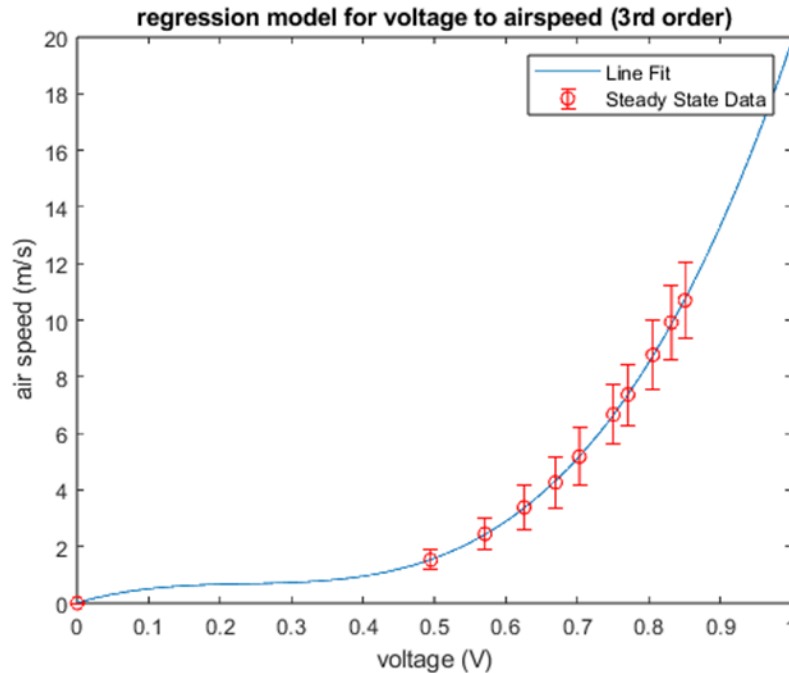


Figure 9: Regression derived from Table 4, steady-state testing. This model shows the relationship between the voltage output of the RevP and a standard airspeed measurement from the Pitot static probe. This model showed some flaws of the wind tunnel tests. The error in the voltage output from the RevP at a speed more incredible than 2 m/s was far outside the specifications the RevP needed to meet; however, the values below 2 m/s were within said specifications, with 0.1 m/s error. This is likely due to vibrations along the RevP thermistor support causing voltage spikes and dips, increasing the overall error at higher speeds.

Upon initial analysis of the RevP's performance, the derived didn't behave like a Hot wire anemometer. Figure 9 shows how the error increases as the airspeed increases in the wind tunnel. Generally, a hot wire anemometer produces a lower error at higher air speeds, and low speeds produce higher errors depending on the application and design of said hot wire anemometer. Due to this information, the senior design team determined that the error was most likely due to the RevP being susceptible to vibrations or fluttering at higher speeds due to its airfoil-like shape and its small size, lacking the rigidity to prevent fluttering in the first place. However, the low airspeed results gave the team the necessary information to move forward with the RevP analysis due to the minor errors produced at steady state conditions in the wind tunnel. RevP calibration would need to be conducted in a test allowing the RevP to move freely to prevent vibrations from interfering with the voltage output.

One limitation of the Arduino MKR was that it could not be powered directly from a 3.7V LiPo battery. This meant a 5V booster was needed to power the microcontroller, barometer,

thermocouple amplifier, IMU, SD card logger, hygrometer, and thermometer. In addition, a 12V booster was required to supply the thermal anemometer with sufficient power (Fig.6). The circuit design was prototyped on a solderless breadboard and verified with a multimeter that 5V and 12V were achieved at proper locations when powered by the 3.7 V LiPo battery. With this information, a custom PCB was designed using KiCad software and then outsourced for production by OshPark (Fig.5). By customizing a PCB, overall weight and footprint reductions were made to make a more thorough and thought-out design.

LoRa functionality was an important feature added to the prototype, allowing for real-time data acquisition and user inputs. The frequency bandwidth is 902 – 930 MHz, allowing multiple channels to communicate with one central receiver to add multiple sondes. The proven transmission range for a LoRa, with a clear line of sight, is 15 km. That given range is well above the necessary range for a tethersonde. Through range testing up to 1km, the LoRa transceiver protocol is sufficient for the tethersonde deployed on a balloon. This testing was completed with only peer-to-peer communication, not a LoRa WAN gateway. A gateway can handle multiple channels at the necessary transmission range.

Testing Methods and Results

After the engineering analysis confirmed that the integration of all the new devices could be done, the senior design team organized multiple tests to create a calibration curve to relate the voltage output of the RevP to windspeed, the response time of the IMU compared to a standard device, and peer to peer data transmission between a fully built tethersonde and a base station.

It was decided that all necessary tests could be completed simultaneously by creating a prototype of a tethersonde with the entire sensor package, deploying it on a testing frame outside to “stress test” the device in a transient environment similar to its actual application. The device that served as the standard was the CSAT3 sonic anemometer. This was provided by the EFD lab and is the standard for windspeed and direction measurements in atmospheric science.



Figure 10: Outdoor testing frame. This lattice mast is outside the EFD lab and includes the CSAT3 on the right and the tethersonde on the left. The tethersonde is supported by a Dyneema cable, much like its actual deployment on a moored balloon. This setup allowed the sonde to rotate freely into the wind, which helped prevent vibrations or fluttering of the RevP thermistor arm.

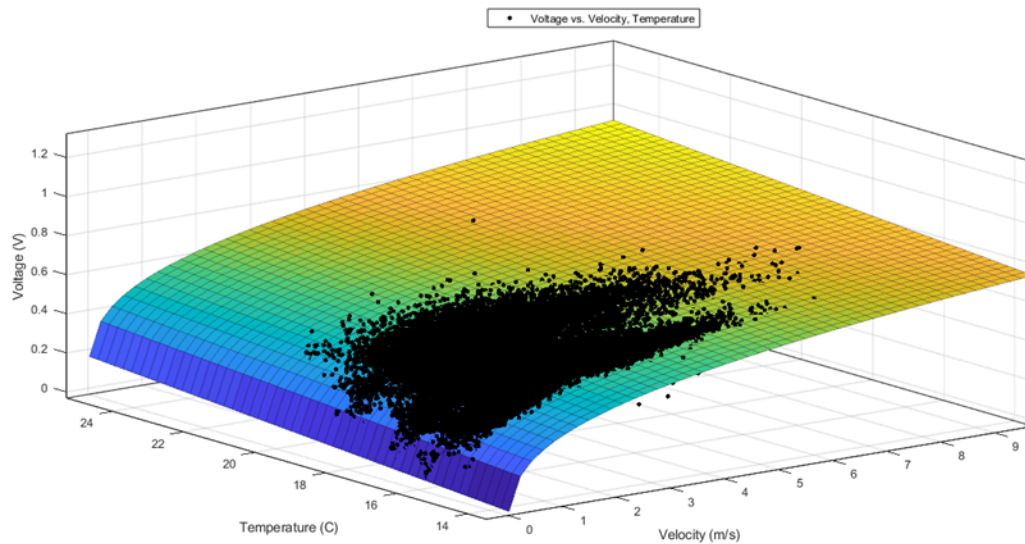


Figure 11: Calibration curve generated by comparing the voltage output of the RevP, along with the air temperature gathered by a local E-type thermocouple, to the velocity magnitude of the sonic anemometer. This model is a power law between voltage and velocity and a linear fit between voltage and temperature (EQ3).

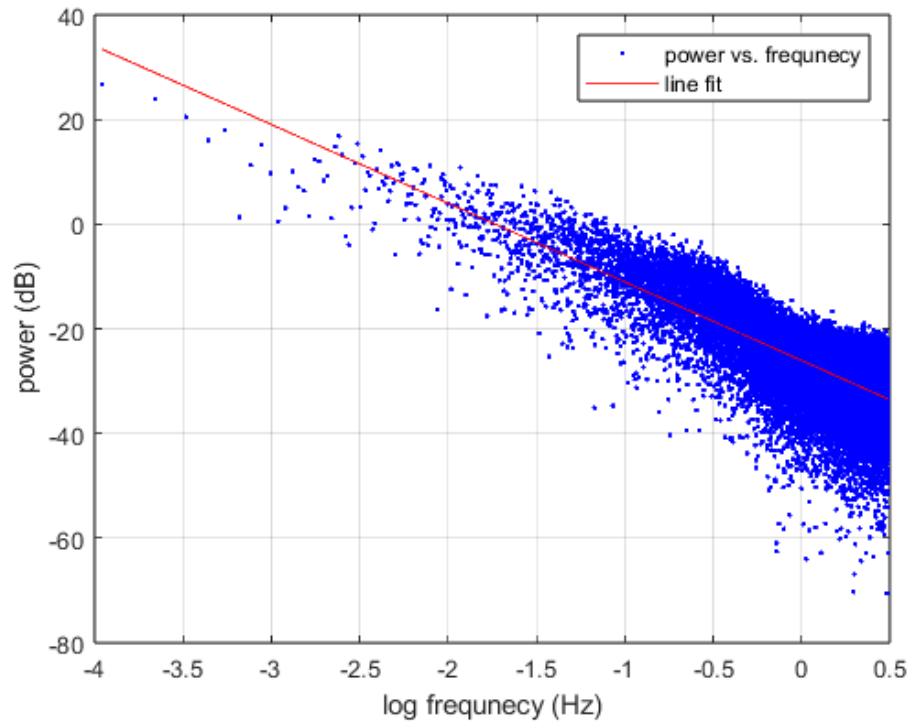


Figure 12: Power spectrum of the RevP voltage output during outdoor testing. This plot shows the alignment with the $-5/3$ decline, a theoretical line that describes turbulent behavior, which indicates turbulent behavior observed in the measurements.

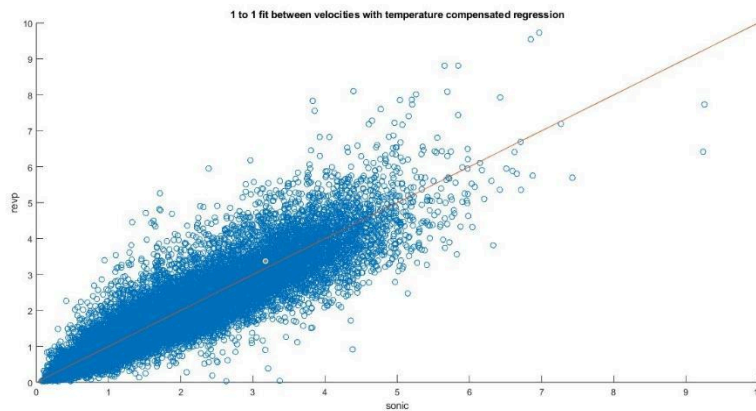


Figure 13: One-to-one plot of the RevP (y-axis) output with the temperature-compensated regression to the Sonic anemometer (x-axis). The alignment along the 1 to 1 line shows a strong correlation between the two data sets. The alignment of these time series was calculated to be ± 0.25 m/s based on the average error (EQ4).

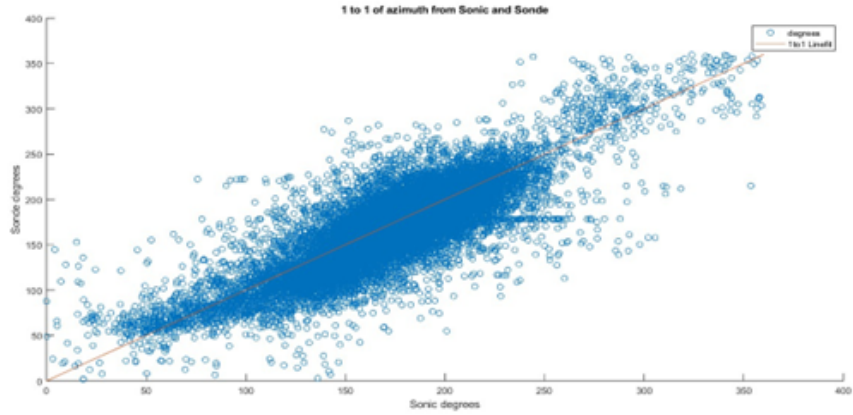


Figure 14: A one-to-one plot of the IMU output (y-axis) to the azimuth produced by the sonic anemometer (x-axis) throughout a 3-hour sonde deployment on the testing frame. The average error made by the difference between these two data sets was 0.433 degrees, exceeding the initial expectations and specifications. This alignment proves that the IMU onboard the sonde is an excellent tool for determining wind direction based on magnetic north values. Magnetic declination had to be accounted for with the Sonic anemometer bearing set to true north (EQ5).

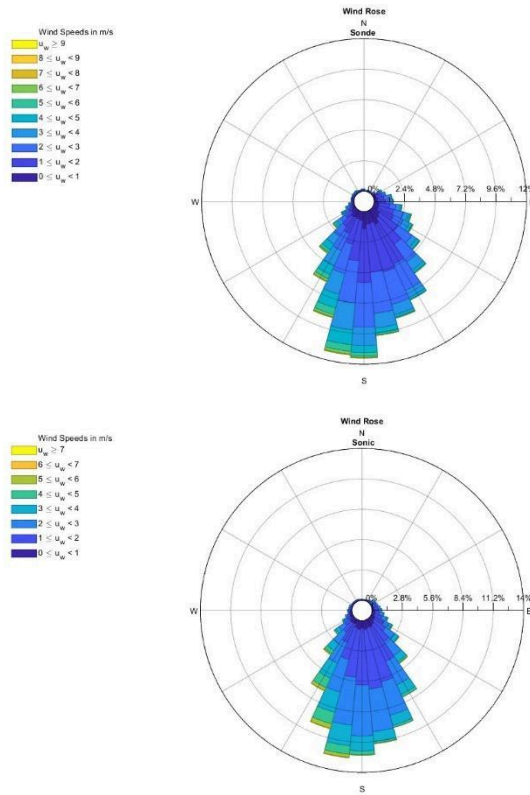


Figure 15: Wind rose comparison of the Sonde (top) and Sonic anemometer (bottom). The wind rose provides air speed and azimuth in a combined figure, which serves as a good marker for the Sonde's performance as a tool for developing a wind profile compared to the profile created by the Sonic anemometer.

The RevP performance was a critical part of this project as there was very little information about the device's performance capabilities. Most of the design process was dedicated to determining the performance of the Rev by first gathering voltage and temperature outputs from a deployed sonde, forming the time-stamped data into a table with the data from the Sonic anemometer gathered at the exact same time. This culminated in the introduction of Figure 11. Utilizing the curve fitting tool in MATLAB, the senior design team could use linear fit against temperature and a power law against the velocity with voltage as the input for both to generate a calibration curve (EQ3). The team determined the average error between the sonde and the sonic from this calibration curve to be 0.25 m/s (EQ4). This velocity error did not meet the user specifications; however, the users were more than surprised by this performance value alone. The RevP by itself is not the ideal anemometer for this application, but with modification to the hot element, this device can be improved drastically. The directional issues and temperature gathering issues were accounted for in the design of the outdoor test (Fig 10) and effectively halved the error from $\pm 0.5 \text{ m/s}$ to $\pm 0.25 \text{ m/s}$ (Prohasky, 2014).

The IMU's performance was another critical part of the Tethersonde development. During the engineering analysis, the preliminary data showed good performance by the Imu in tracking a north bearing; however, the alignment shown in Figure 14 depicts the IMU's response to wind direction, and its alignment to all cardinal directions it was aligned in is more than sufficient than at a 10-degree error. The IMU exceeded expectations for a direction-gathering tool.

For the critical functions of the tethersonde, wind speed and direction, including an IMU and the RevP, proved to be good fits for the sensor package. The alignment wind profiling between the Sonic and the tethersonde shows that the Sonde, as it stands now, is well on its way to becoming a powerful atmospheric data acquisition tool. Figure 13 shows how the calibration curve can produce well-aligned velocity values with that of a standard device, but Figure 12 shows that the thermal time constant of the thermistor is higher than what is needed for a fast-response hot wire anemometer. One of the primary goals of this project was to keep the sensor package low-cost so the RevP should not be gone due to its performance. The RevP can be improved by removing the thermistor and using a platinum wire as the heated element. This will lower the thermal time constant and improve the shape factor, which helps gather velocity magnitudes during a wind shift.

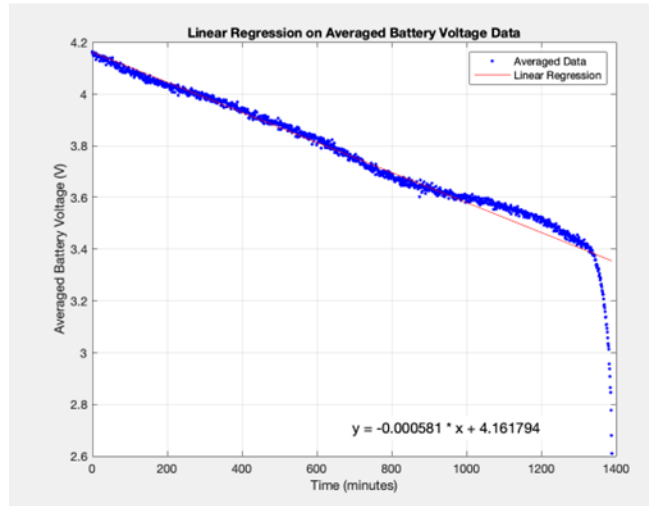


Figure 16: Battery discharge test with a 3500 mAh battery included in the tethersonde. This test determined the low voltage value that would act as the threshold for when to power off the tethersonde and replace the battery.

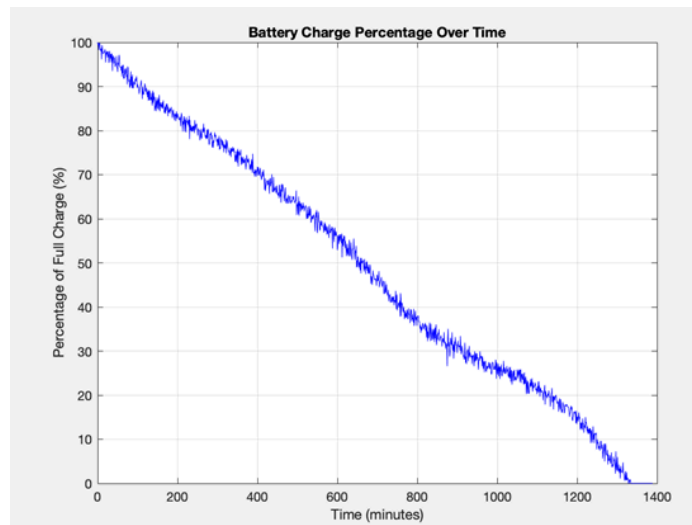


Figure 17: Adjusted battery discharge mod, which scales the low threshold to 0% battery. This is embedded within the tethersonde microcontroller to blink red when the 0% value is reached.

Additional features, such as battery indication and radio communication, were successfully added to the Sonde. A battery discharge model (Fig 16) was created so the team could find the low voltage threshold the tethersonde would need to know to stop logging data (Fig 17). LoRa connectivity was very successful for peer-to-peer communication at 5Hz. All

initial data acquisition was achieved through the built-in LoRa chip on each microcontroller. Data could be transmitted from the sonde to another Arduino connected to a MATLAB script, which allowed for real-time data acquisition and plotting.

Early in the project, the team developed the reach goal of building a tethersonde base station that could handle multiple tethersonde transmissions. According to the LoRaWAN network server information, the proper hardware is called a LoRaWAN gateway. The team purchased a Raspberry Pi gateway due to its customizability and low power requirements. This would be the perfect tool for a remote system to handle all the incoming messages; however, all the information regarding the programming involved for the gateway is related to the “Internet of Things” system, which for a remote system like the tethersonde is not ideal.

The team explored creating a fully local network server, but the latency of messages that could be collected was more than 30 seconds per message no matter what the Sonde transmission rate was. With information being lost in the system and the latency of messages, the base station remains a dedicated Arduino MKRWAN1310 with an increased SPI frequency to handle a frequency cycling program so that it can be received on multiple frequencies. This is not an ideal base station. For the researchers who will be taking over the project from the senior design team, it is recommended that they explore rebuilding the Raspberry Pi gateway by utilizing MATLAB's built-in Raspberry Pi interface toolbox to decode the Lora packets and acquire tethersonde data at an appropriate rate.

Comparison of Design Metrics

Table 5: The milestone for the ‘complete closed-loop wind tunnel test’ was excluded because the responsible faculty member was, and currently is, abroad conducting fieldwork.

Milestone	Date	Rev. Date	Status
Complete closed loop wind tunnel tests	02-02	NA	Excluded
Compare thermal anemometer in 2D setup to 3D sonic anemometer outside.	02-02		Complete
Add existing components from the original prototype	02-02		Complete
Compile code into the arduino script as well as modify the matlab script to accept the data stream	02-02		Complete
Design and create PCB for prototype	02-02		Complete
Create housing design that will fit the PCB and any external sensors.	02-16		Complete

Appendix

EQ1: King's law heat transfer

$$I^2 R_w = \frac{V^2}{R_w} = (T_a - T_w)(a + bU^c)$$

EQ2: Modified Bernoulli's equation

$$U = \sqrt{\frac{\rho(\Delta P + P_b)}{2}}$$

EQ3: Regression model

$$V = a + bT_w + cU^d$$

EQ4: average error

$$Err_{avg} = \frac{1}{n} \sum \left(\frac{|A-M|}{A} \right)$$

EQ7: Magnetic declination

$$\theta_{true} = \theta_{mag} + \theta_{dec}$$